Belief-Aware Deep Multi-Agent Reinforcement Learning for Decentralized Visual Sensor Networks

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Abstract. Deep sensor networks require multiple sensing agents to coordinate their behaviors from local observations while adapting to changes in target motion, sensing conditions, and interactions among neighboring sensors. Such decentralized coordination is challenging because each sensor has only partial information about the environment and cannot directly observe the evolving behaviors of its teammates, leading to non-stationary learning dynamics and poor coordination. We propose Belief-RAC, an uncertainty-aware decentralized multi-agent reinforcement learning method for adaptive coordination in deep sensor networks. Belief-RAC represents changing local interaction patterns as latent contexts and infers them online through a causal categorical belief model using only the current local observation, previous action, and previous shared reward. Without requiring ground-truth context labels, the belief model is trained self-supervisedly to predict context-dependent observation transitions and rewards. Based on the inferred belief, each sensor combines posterior-weighted context values with an optimistic estimate across possible contexts, while dynamically adjusting the degree of optimism according to belief uncertainty and detected interaction changes. This design allows sensors to exploit reliable contextual information while maintaining sufficient exploration of potentially high-value cooperative behaviors, thereby alleviating both non-stationarity and relative overgeneralization. Experiments on cooperative target coverage in the Multi-Agent Tracking Environment show that Belief-RAC improves coordination performance and adapts effectively to changing interaction dynamics in decentralized sensor networks.

Keywords: Cooperative Multi-Agent Reinforcement Learning · Fully Decentralized Learning · Multi-Agent Tracking · Deep Reinforcement Learning

1 Introduction

Cooperative multi-agent reinforcement learning (MARL) provides a natural framework for distributed sensor networks, where multiple sensing agents must coordinate their sensing and decision-making to achieve a shared objective [22,9]. This paradigm has been increasingly studied for cooperative sensing tasks such as target coverage, localization, search, and multi-target tracking [25,1,24]. In these settings, each sensor can only see a small part of the environment, and its observations may not always be correct. To track targets well, sensors need to work together. However, this is difficult because each sensor has limited information, target movements are hard to predict, and the actions of other sensors keep changing.

Cooperative MARL generally uses one of two approaches: centralized training with decentralized execution (CTDE) and decentralized training and execution (DTE) [2]. CTDE leverage global data while learning operate independently using only local information during deployment, including methods such as VDN [19], QMIX [16], and MAPPO [23]. DTE training and execute agents completely independently, meaning agents only use their own local information during both phases. This lack of reliance on centralized coordination makes DTE especially well-suited to settings such as distributed sensor networks, where constraints on communication bandwidth, energy, and real-time responsiveness often render centralized coordination infeasible. Fully decentralized MARL has been applied to distributed vehicle tracking in wireless sensor networks [13], while distributed multi-agent learning has also been investigated for routing and control in resource-constrained underwater and mobile sensor networks [12,26]. Related studies in directional sensor networks further highlight the need for scalable coordination as the number of sensing agents and sensor-target interactions increases [22,17].Independent methods such as IQL [20] and IPPO [21] demonstrate that effective cooperative behavior can emerge without centralized training. However, DTE remains challenging because concurrently changing teammate policies induce *non-stationarity*, while *relative overgeneralization* can cause promising cooperative actions to be undervalued when paired with exploratory or suboptimal teammate behavior [8]. Recent approaches address these difficulties from different perspectives. I2Q reduces non-stationarity through idealized transition dynamics [6], while TVPO develops decentralized policy optimization with theoretical convergence guarantees [18]. BQL introduces a best-possible operator to improve learning under changing effective transition dynamics [7]. More recently, RAC models variations in teammate behavior as contexts and combines context-conditioned values with optimistic marginalization to jointly alleviate non-stationarity and relative overgeneralization [10]. Nevertheless, reliably inferring the current interaction regime from strictly local information while controlling optimism under uncertainty remains challenging.

Figure 1 illustrates the gap that motivates our approach: a context assigned from the completed episodic return is available only at the end of each episode, while a sensor must act throughout the episode. Therefore motivated the online

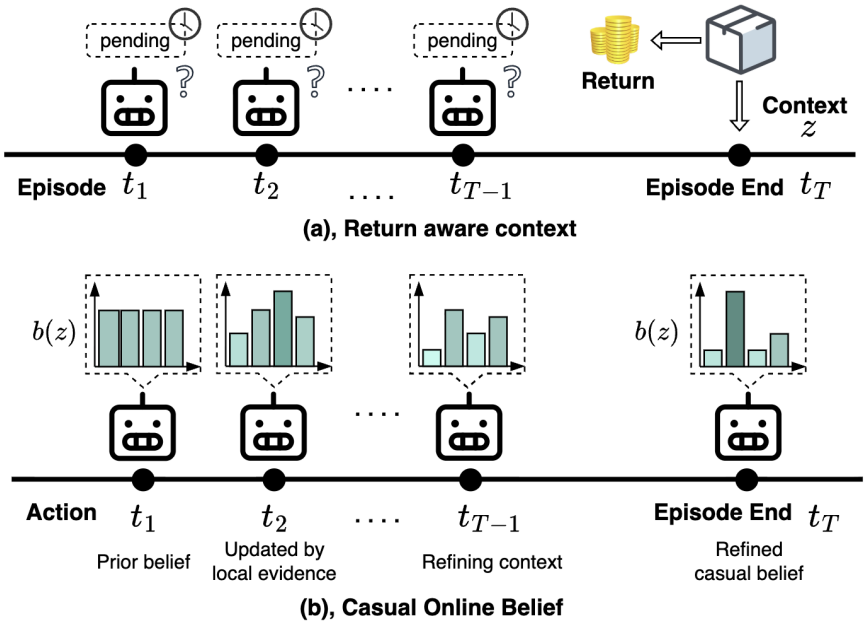


Fig. 1: **(a)** RAC sets context z based on the total reward of an episode, so it is only available after the episode ends. **(b)** Each agent's belief is updated at every step using its local information only.

belief using locally available evidence to inform each decision without waiting for the episode to end.

We propose BELIEF-RAC, an online belief-aware extension of Return-Aware Context (RAC) [10] for cooperative decision-making in deep sensor networks. Each sensor agent learns from local observations, while changes in other sensors's policies make its effective sensing task non-stationary. RAC represents these changes through contexts derived from episodic returns, which are unavailable at decision time. BELIEF-RAC instead causally infers a categorical belief over coordination contexts from the current local observation, previous action, and previous shared reward. Transition and reward prediction trains the belief model using self-supervised learning, while an uncertainty adaptive rule combines posterior expectation and optimistic context selection. This enables online adaptation to changing sensor coordination while preserving fully decentralized execution. Our contributions can be summarized as follow:

- We propose a method that uses local information to estimate hidden coordination contexts and learns without context labels by predicting how the environment changes.

- We introduce a belief-aware value aggregation rule that adjusts its optimism based on belief uncertainty and changes in coordination context, which handle both non-stationarity relative overgeneralization.
- We evaluate Belief-RAC against DTE baselines in the Multi-Agent Tracking Environment (MATE) [15], a representative cooperative tracking benchmark for distributed sensor networks.

2 Preliminaries

2.1 Multi-Agent Markov Decision Process

We formulate the cooperative task as a Multi-Agent Markov Decision Process (MMDP) [4], defined by the tuple $\mathcal{M} = \langle \mathcal{N}, \mathcal{S}, \mathcal{A}, P, R, \gamma \rangle$ where $\mathcal{N} = 1, \ldots, n$ is the set of n agents, $\mathcal{S}$ is the global state space, and $\mathcal{A} = \mathcal{A}^1 \times \cdots \times \mathcal{A}^n$ is the joint action space, here $\mathcal{A}^i$ denotes the action space of agent i, transition function $P(\cdot \mid s_t, \boldsymbol{a}_t)$ gives the probability of transitioning to next state after executing the joint action $\boldsymbol{a}_t$ at state s_t, while $r_t = R(s_t, \boldsymbol{a}_t)$ defines the shared reward received by all agents, $\gamma \in [0, 1)$ is the discount factor. At each time step t, every agent i observes the global state s_t and selects an action a_t^i according to its policy $\pi^i(a_t^i \mid s_t)$. The individual actions form the joint action $\boldsymbol{a}_t = (a_t^1, \ldots, a_t^n)$. The objective is to learn a joint policy $\boldsymbol{\pi} = (\pi^1, \ldots, \pi^n)$ that maximizes the expected cumulative discounted return, $\mathbb{E}_{\boldsymbol{\pi}, P}\left[\sum_{t=0}^{\infty} \gamma^t r_t\right]$.

In the sensor-network setting, each sensor has access only to its local information, not the complete global state, so this problem can be described by the Dec-POMDP [3] $\mathcal{M} = \langle \mathcal{N}, \mathcal{S}, \{\mathcal{A}^i\}_{i \in \mathcal{N}}, \{\mathcal{O}^i\}_{i \in \mathcal{N}}, P, R, Z, \gamma \rangle$. Here, $\mathcal{O}^i$ denotes the local observation space of sensor i, $o_t^i \in \mathcal{O}^i$ is its observation at time t, Z is the model characterizes how local observations are generated from the underlying environment state and joint action. A sensor i selects its action according to a decentralized policy $\pi^i(a_t^i \mid o_t^i)$ using only its locally available observation. The joint action $\boldsymbol{a}_t$ determines the subsequent state according to P, while the common reward R encourages the sensors to cooperate toward the target coverage objective. The overall goal remains to optimize the joint policy $\boldsymbol{\pi}$ so as to maximize the expected discounted team return.

2.2 Contextual MDP and Return-Aware Context

A contextual MDP represents a family of tasks indexed by a context $c \in \mathcal{C}$, where $\mathcal{C}$ is the context space [5]. Here, we specialize this formulation to the local, per-agent view induced by the Dec-POMDP above, fixing an agent $i \in \mathcal{N}$ rather than defining a new joint multi-agent model. The mapping $M(c) = \langle \mathcal{S}, \mathcal{A}^i, P_c^i, R_c^i, \gamma \rangle$ assigns each context a local MDP with context-specific transitions P_c^i and rewards R_c^i, while retaining the same state and action spaces. In decentralized MARL, contexts can represent distinct teammate-policy regimes [10,11].

RAC [10] uses the completed episodic return $G = \sum_{t=0}^{T-1} r_t$, with episode length T, as a proxy for the coordination regime. RAC maps the completed

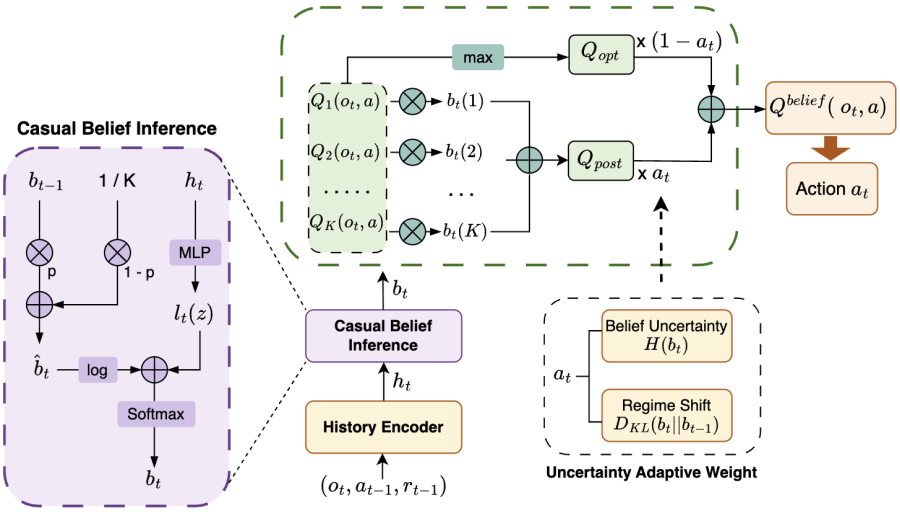


Fig. 2: BELIEF-RAC overview: local history informs online context beliefs, while uncertainty-adaptive value aggregation guides decentralized action selection.

episodic return G to a discrete context label $c \in \mathcal{C} = 1, \ldots, K$. Specifically, the range $[G_{\min}, G_{\max}]$ is divided into K intervals. The context is determined by the interval containing G,then it is represented as a K-dimensional one-hot vector, with the entry corresponding to the selected interval set to one. This context label is used as an additional input to the local action-value function $Q^i(\tau_t^i, c, a)$: the estimated discounted return for candidate action $a \in \mathcal{A}^i$, given context c [10] and local history $\tau_t^i = (o_0^i, a_0^i, o_1^i, a_1^i, \ldots, a_{t-1}^i, o_t^i)$. RAC reduce relative overgeneralization by using the optimistic marginal value $Q_{\text{opt}}^i(\tau_t^i, a) = \max_{c \in \mathcal{C}} Q^i(\tau_t^i, c, a)$. However, the return-defined context is unavailable before an episode ends. This motivates BELIEF-RAC, which infers contexts online and adapts optimism to belief uncertainty. Unlike RAC's return-derived context c, BELIEF-RAC uses a latent context z, whose probability distribution is inferred online from local evidence.

3 Methodology

3.1 Overview

Fully decentralized cooperative MARL is challenging because each agent must adapt to the behavior of its teammates without observing their actions, policies, or internal states. From the perspective of a single agent, changes in teammate policies therefore appear as changes in the effective environment. RAC [10] addresses this problem by introducing discrete coordination contexts and learning context-conditioned action values. It additionally evaluates an action optimistically across

contexts, preventing potentially cooperative actions from being suppressed by exploratory or incompatible teammate behavior. However, RAC [10] identifies the context of an episode from its completed episodic return. The context is therefore useful for organizing training experience, but is unavailable when an action must actually be selected.

RAC can nevertheless act online through a context-free main branch trained by TD learning and supervision from optimistic context values. We propose Belief-RAC, which replaces retrospective context assignment with an online belief over latent coordination contexts z. Figure 2 summarizes belief inference and value aggregation. For each agent, we maintain

$$b_t^i(z) = \Pr(z_t^i = z \mid \tau_t^i), \qquad z \in \{1, \dots, K\}, \tag{1}$$

where τ_t^i denotes the locally available history before action a_t^i is selected, z_t^i denotes the latent coordination context of agent i at time step t, and K is the number of latent coordination contexts. These contexts are not provided by the environment and do not correspond to explicit teammate actions. Instead, they are learned as latent regimes that explain different local transition and reward patterns, which may reflect target motion and observation variability as well as changing teammate behavior. A recurrent model estimates b_t^i using local information and learns by predicting the next state and reward. To choose actions, it combines context values weighted by the belief with the highest context value.

3.2 Online Belief over Coordination Contexts

Before acting at time t, agent i forms a local evidence vector

$$\xi_t^i = \left[o_t^i,\, \mathrm{onehot}(a_{t-1}^i),\, r_{t-1},\, \mathrm{id}^i\right],$$

where o_t^i is the local observation at timestep t received by agent i, $\mathrm{onehot}(a_{t-1}^i)$ is the one-hot encoding of the previous action, r_{t-1} is the shared reward received at the previous time step, and id^i is an optional encoding of the agent identity. We initialize $b_{-1}^i(z) = 1/K$, yielding $\bar{b}_0^i(z) = 1/K$, and set the previous action and reward inputs to zero at episode start. A recurrent network receives the local input sequence and produces logits $\ell_t^i(z)$ for each latent context. We combine these logits with a predictive prior derived from the previous posterior:

$$\bar{b}_t^i(z) = \rho b_{t-1}^i(z) + \frac{1-\rho}{K}, \quad b_t^i = \mathrm{softmax}_z\big(\ell_t^i(z) + \beta \log \bar{b}_t^i(z)\big). \tag{2}$$

Here, $0 \le \rho \le 1$ and $\beta \ge 0$. The recurrent network uses local information over time to produce context scores. The prior is predicted from the previous context belief. We then combine the scores with the prior using Bayes' rule to update the belief, where $\bar{b}_t^i(z)$ acts as the prior distribution and $\ell_t^i(z)$ can be considered as the likelihood.

3.3 Self-Supervised Discovery of Latent Contexts

Without ground-truth context labels, we train the belief filter to predict local observation changes and rewards. We project the observation difference using a fixed random matrix $P_{\text{proj}} \in \mathbb{R}^{d_p \times d_o}$, where d_o is the local observation dimension and $d_p = \min(d_o, 16)$ in our experiments. The projection matrix P_{proj} is initialized by drawing i.i.d. samples $X_{jk} \sim \mathcal{N}(0,1)$ and ℓ_2-normalizing each row, yielding $[P_{\text{proj}}]_{j,:} = X_{j,:}/\|X_{j,:}\|_2$. We then define the projected observation target as $\Delta\widetilde{o}_t^i = P_{\text{proj}}(o_{t+1}^i - o_t^i) \in \mathbb{R}^{d_p}$. Then a mixture decoder takes $(o_t^i, a_t^i, \text{id}^i)$ as input, for each context z it predicts both the projected observation change and the team reward. For context z, its prediction error is

$$\epsilon_t^i(z) = \frac{1}{d_p} \sum_{j=1}^{d_p} \ell_{\text{SL1}}\left([\widehat{\Delta\widetilde{o}}_t^i(z)]_j - [\Delta\widetilde{o}_t^i]_j \right) + \lambda_r \ell_{\text{SL1}}\left(\hat{r}_t^i(z) - r_t\right), \tag{3}$$

where $\widehat{\Delta\widetilde{o}}_t^i(z) \in \mathbb{R}^{d_p}$, $\hat{r}_t^i(z)$ are the decoder's context-specific predictions, ℓ_{SL1} is the smooth-L_1 penalty, and $[\cdot]_j$ selects the jth projected coordinate. The first term averages the transition prediction error over the d_p coordinates, while $\lambda_r \geq 0$ weights the scalar reward prediction error. To encourage different context slots to specialize, each transition is assigned to a single context during reconstruction. We draw a hard straight-through Gumbel-Softmax assignment $\widetilde{z}_t^i$ using the posterior logits $g_t^i(z) = \ell_t^i(z) + \beta \log \bar{b}_t^i(z)$, which combine the recurrent evidence with the predictive prior. The forward pass therefore uses a discrete context, while the straight-through estimator allows the reconstruction signal to train the evidence encoder. The belief-learning objective is

$$\mathcal{L}_{\text{belief}} = \frac{1}{|\mathcal{M}|} \sum_{(i,t)\in\mathcal{M}} \left[\epsilon_t^i(\widetilde{z}_t^i) + \lambda_{\text{kl}} \text{KL}(b_t^i \| \mathcal{U}_K)\right] + \lambda_{\text{bal}} \text{KL}(\bar{b}_{\mathcal{M}} \| \mathcal{U}_K), \tag{4}$$

Here, $\mathcal{M}$ denotes the valid non-padded transitions, $\epsilon_t^i(\widetilde{z}_t^i)$ is the context-specific prediction error, $\mathcal{U}_K(z) = 1/K$ is the uniform context prior, and $\bar{b}_{\mathcal{M}}$ is the batch-averaged posterior. We employ two KL divergence terms to regularize context usage and avoid mode collapse. The term $\text{KL}(b_t^i \| \mathcal{U}_K), \lambda_{\text{kl}} \geq 0$, penalizes overly concentrated individual posteriors. The batch-level term $\text{KL}(\bar{b}_{\mathcal{M}} \| \mathcal{U}_K), \lambda_{\text{bal}} \geq 0$, encourages balanced slot utilization across the minibatch. Together, these terms counterbalance the prediction objective's tendency toward context specialization.

Following RAC [10], each agent maintains a context-conditioned value function $Q_{\text{twin}}^i(\tau_t^i, z, a)$. The agent's local history is encoded once into a shared representation $h_{\tau,t}^i$. A small hyper-network maps the one-hot representation of each context z to the parameters of the final action-value layer:

$$(W_z, v_z) = \text{Hyper}(\mathbf{e}_z), \qquad Q_{\text{twin}}^i(\tau_t^i, z, a) = \left[W_z h_{\tau,t}^i + v_z\right]_a, \tag{5}$$

where $\mathbf{e}_z$ is the one-hot encoding of context z, W_z and v_z are the generated weights and bias, and $[\cdot]_a$ selects the component for action a. All K context heads reuse $h_{\tau,t}^i$, requiring only one history encoding. RAC associates transitions with a

context obtained from the completed episodic return. BELIEF-RAC instead uses the causal hard assignment $\widetilde{z}_t^i$ inferred from the online belief model to evaluate $Q^i_{\text{twin}}(\tau_t^i, \widetilde{z}_t^i, a_t^i)$ during training. The context slots learned in this way are the same slots evaluated and aggregated under the full posterior during execution. We apply double Q-learning to the context-conditioned branch. For each sampled transition, the next action is selected using the online twin value function as $a_{t+1}^{i,*} = \arg\max_{a \in \mathcal{A}_{t+1}^i} Q^i_{\text{twin}}(\tau_{t+1}^i, \widetilde{z}_{t+1}^i, a)$. The corresponding TD target is then computed with the target network as $y^i_{t,\text{twin}} = r_t + \gamma(1-d_t) Q^{i,-}_{\text{twin}}(\tau_{t+1}^i, \widetilde{z}_{t+1}^i, a_{t+1}^{i,*})$. In both action selection and target evaluation, we condition on the next-step context $\widetilde{z}_{t+1}^i$. The corresponding value loss is the squared TD error between $y^i_{t,\text{twin}}$ and $Q^i_{\text{twin}}(\tau_t^i, \widetilde{z}_t^i, a_t^i)$. Here, $\mathcal{A}_{t+1}^i$ contains the available next-step actions, the superscript $-$ denotes the target network, and d_t indicates episode termination. The next context is inferred from o_{t+1}^i, a_t^i, r_t, and the preceding local history, all available before selecting a_{t+1}^i. This remains causal and does not assume that the context is unchanged across a transition.

3.4 Joint Learning and Fully Decentralized Execution

For each action, we compute the posterior-weighted value as $Q^i_{\text{post}}(\tau_t^i, a) = \sum_z b_t^i(z) Q^i_{\text{twin}}(\tau_t^i, z, a)$. The optimistic value is the maximum across contexts, given by $Q^i_{\text{opt}}(\tau_t^i, a) = \max_z Q^i_{\text{twin}}(\tau_t^i, z, a)$. Q_{post} averages action values across contexts, Q_{opt} selects the highest predicted value for each action across contexts. This preserves high values that posterior averaging may reduce. We measure the discrepancy between the predictive prior and posterior by $s_t^i = \text{JSD}_2(b_t^i \| \bar{b}_t^i) \in [0,1]$ using the Jensen-Shannon divergence [14] computed with base-2 logarithms between the predicted and updated beliefs, then compute uncertainty as $u_t^i = 1 - \big(1 - H_{\text{norm}}(b_t^i)\big)(1 - s_t^i)$ and set the optimism coefficient to $\alpha_t^i = \alpha_{\min} + (\alpha_{\max} - \alpha_{\min}) u_t^i$, where $H_{\text{norm}}(b_t^i) \in [0,1]$ is the categorical entropy normalized by $\log K$, $0 \le \alpha_{\min} \le \alpha_{\max} \le 1$. The decision value is the linear combination of the posterior and optimistic estimates:

$$Q^i_{\text{dec}}(\tau_t^i, a) = (1 - \alpha_t^i) Q^i_{\text{post}}(\tau_t^i, a) + \alpha_t^i Q^i_{\text{opt}}(\tau_t^i, a). \tag{6}$$

The belief filter and context-conditioned value function learn from locally available transitions sampled from replay. For agent i, $\mathbb{E}_{D^i}$ denotes the minibatch average over valid transitions $(\tau_t^i, a_t^i, r_t, \tau_{t+1}^i, d_t)$ in D^i, excluding padded steps and steps after termination. Using the Double Q target $y^i_{t,\text{twin}}$ defined above, the value-learning objective is

$$\mathcal{L}^{\text{twin}}_{\text{TD}} = \mathbb{E}_{D^i}\left[\left(y^i_{t,\text{twin}} - Q^i_{\text{twin}}(\tau_t^i, \widetilde{z}_t^i, a_t^i)\right)^2\right], \tag{7}$$

The objective for jointly learning the belief and context-conditioned value components is

$$\mathcal{L}_{\text{core}} = \mathcal{L}^{\text{twin}}_{\text{TD}} + \lambda_{\text{belief}} \mathcal{L}_{\text{belief}}, \qquad \lambda_{\text{belief}} \ge 0. \tag{8}$$

The TD objective learns context-dependent action values. Its gradients also train the belief model through the straight-through context assignments.

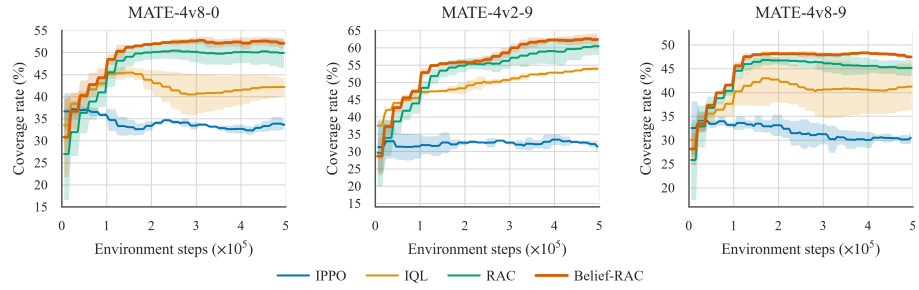
4 Experiments

4.1 Experimental Settings

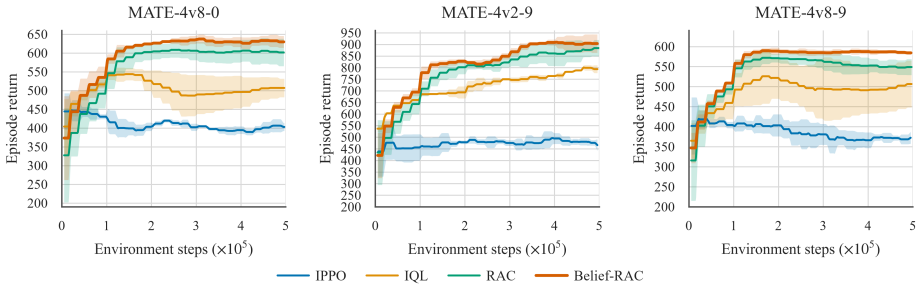
Environment and tasks. We conduct our main experiments in the Multi-Agent Tracking Environment (MATE) [15], where a team of camera agents cooperatively tracks moving targets using local observations. We consider three scenarios: MATE-4v8-9, MATE-4v8-0, and MATE-4v2-9, containing four cameras with eight targets and nine obstacles, eight targets and no obstacles, and two targets and nine obstacles, respectively. The main comparisons use the supplied run logs for each scenario, with 5×10^5 environment steps per run.
Compared methods. We compare BELIEF-RAC against IPPO [21], IQL [20], and RAC [10].
Network architectures. All methods process each agent's observations using recurrent networks. IPPO employs separate actor and critic networks, each comprising a 512-256 tanh MLP, a 256-unit GRU, and a linear head for action-masked logits or scalar values. IQL, RAC, and Belief-RAC use the same Entity-Attention-GRU architecture. Camera, target, and obstacle features are encoded via type-specific 128-128 SiLU MLPs. External entities are aggregated using a visibility-masked, four-head attention mechanism with the camera as the query. These features are fused into a 256-dimensional representation using SiLU and LayerNorm and passed to a 256-unit GRU, along with a 32-dim previous-action embedding. IQL uses a single linear Q-head. The context-conditioned value networks in RAC and Belief-RAC use hypernetwork-generated heads to evaluate $K = 10$ contexts.

4.2 Main Results

We compare test performance using coverage rate and episode return across the three MATE scenarios. Figure 3 presents test evaluations over 5×10^5 environment steps, with three random seeds per method and scenario. Higher values indicate better performance for both metrics.
Test performance. BELIEF-RAC consistently outperforms IPPO, IQL, and RAC in mean final test coverage and episode return across all three MATE scenarios, with results averaged over three seeds. Belief-RAC not only achieves higher final returns but also learns much more stably across different seeds than IQL. It also coverging faster on MATE-4v8-0 and MATE-4v8-9 with coverage and return stabilize at roughly 1.5×10^5 steps. Although learning procedure of BELIEF-RAC continues more slowly on MATE-4v2-9, its coverage rate and episodic return is clearly higher. IQL consistently underperforms the context-based methods by the end of training. IPPO reamains struggle to maintain late-stage improvements. Comparing with the previous about contextual MARL, Belief-RAC consistently beats its original motivated method-RAC. Since both share the identical value backbone, this gap directly validates the effectiveness of our belief-based context inference and adaptive value aggregation.



(a) Test coverage rate.



(b) Test episode return.

Fig. 3: Test performance on MATE-4v8-0, MATE-4v2-9, and MATE-4v8-9 (left to right). Each panel compares IPPO, IQL, RAC, and BELIEF-RAC under the same environment-step budget.

4.3 Ablation Study

We compare five variants of textscBelief-RAC on MATE-4v8-9 under a 5×10^5 step training budget, each remove a components of the full method. All other settings remain unchanged. We compare them in 3 runs with 3 different seeds.

Without belief auxiliary loss. We removing the reconstruction objective and both belief regularizers from optimization by set $\lambda_{\text{belief}} = 0$ in Eq. (8). The belief network remains active and continues to receive gradients from the context-conditioned TD objective through the straight-through context assignments. This variant tests whether auxiliary supervision improves context learning beyond the reinforcement-learning signal.

Without temporal prior. We removes the predictive prior's contribution to the posterior update by set $\beta = 0$ in Eq. (2), yielding $b_t^i = \text{softmax}(\ell_t^i)$. The predictive prior is still computed for the shift score. This variant therefore tests the benefit of explicit categorical prior integration beyond recurrent evidence encoding.

Without adaptive optimism. By set $\alpha_{\min} = \alpha_{\max} = 0.5$ in Eq. (6), we got $Q_{\text{dec}}^i = 0.5Q_{\text{post}}^i + 0.5Q_{\text{opt}}^i$. We use this fixed weight for both the decentralized

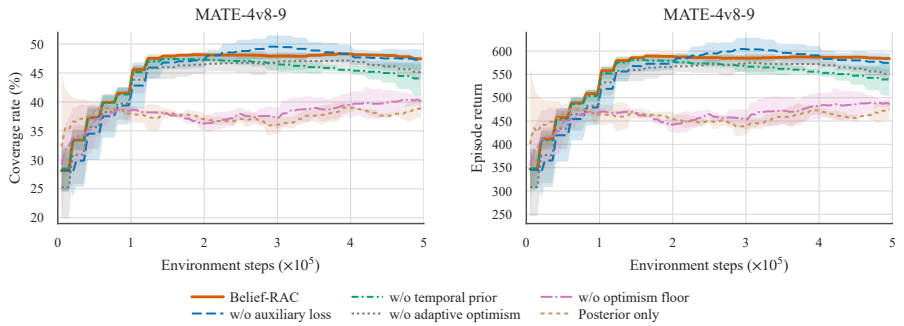


Fig. 4: Ablation results on MATE-4v8-9. The full BELIEF-RAC model is compared with variants without auxiliary belief loss, without temporal prior integration, with fixed optimism ($\alpha = 0.5$), without an optimism floor ($\alpha = u$), and with posterior-only value aggregation ($\alpha = 0$).

network during training and action selection during execution. This variant keeps both posterior-based and optimistic values, but always gives them equal weight. It tests whether adjusting the weight based on uncertainty performs better than using a fixed weight.

Without optimism floor. We set $\alpha_{\min} = 0$ and $\alpha_{\max} = 1$ in Eq. (6), yielding $\alpha_t^i = u_t^i$. When belief uncertainty is low, the decision value can approach the posterior weighted estimate without retaining a minimum level of optimism. We want to know whether preserving a positive optimism floor improves coordination compared with weighting optimism directly by uncertainty.

Posterior policy only. This removes the optimistic contribution by set $\alpha_{\min} = \alpha_{\max} = 0$ in Eq. (6), which means $Q_{\text{dec}}^i = Q_{\text{post}}^i$. . Action selection during training rollouts and execution uses only the posterior weighted context values. We investigated whether optimistic aggregation improves coordination beyond posterior based value estimation only.

Results. The full BELIEF-RAC method provides the strongest overall combination of mean performance and consistency across seeds. It finishes with the highest mean coverage and episode return, while its narrower late-training standard-deviation bands show the lowest across-seed variability among the compared variants. The component comparisons indicate different roles. Removing auxiliary belief supervision leaves the final means close to those of the full method but the stability of training across 3 runs decreased, suggesting that its main benefit is more consistent learning between different instances of the environment. Removing the temporal prior and replacing adaptive optimism with a fixed weight produces lower final means and showing a late-training decline. The most significance decreases in both coverage and return occur for posterior policy only and without optimism floor. These variants respectively remove the optimistic contribution entirely and remove its positive lower bound. Their larger performance losses suggest that retaining optimism and a positive optimism floor

are the two most influential components in these comparisons. This ranking is descriptive, subject to the configuration limits stated above; it does not establish statistical significance.

5 Conclusion

We presented Belief-RAC, a decentralized cooperative MARL method that replaces retrospective return-based context assignment with online latent-context inference. A recurrent belief filter is trained with context-conditioned TD learning and auxiliary transition and reward prediction. A belief-aware decision rule combines posterior-weighted and optimistic context values using posterior entropy and prior-posterior discrepancy. Across three MATE scenarios, the method achieves higher mean final test coverage and episode return than IPPO, IQL, and RAC. The largest observed ablation losses on MATE-4v8-9 occur when removing optimism or its positive floor, suggesting that these are the two most influential components in the available comparisons.

Limitations. First, the learned contexts may capture not only teammate behavior, but also changes in target motion, random events in the environment, or noisy observations. This makes the meaning of each context harder to interpret. Second, our uncertainty measures are heuristic. They can be useful in practice, but they do not guarantee that the belief is well calibrated or that every change in behavior is detected correctly. Finally, the number of contexts and the context transition rules are fixed in advance. This can reduce flexibility, and the belief modules also add extra computation

Future work. We will test how the learned contexts change when teammate policies are changed in a controlled way and measure how well the predicted beliefs match the true outcomes. We also plan to improve efficiency and flexibility by allowing the model to adjust the number of context slots, learning context transitions from data. We plan to extend this method to other multi-agent reinforcement learning environments, such as StarCraft Multi-Agent Challenge (SMAC).

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